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## 2026 QSR Webinar Series

# Robust Inference of Unseen Heterogeneous Physical Properties from Imperfect Data: A Unified Probabilistic Physics-Informed Machine Learning Framework




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**Zoom ID Registration Link: https://njit-edu.zoom.us/meeting/register/W4EaHllRQhG1Xe1X8ivsMg**


**Abstract:** Many biomedical and engineering systems are characterized by spatially heterogeneous physical properties that govern their responses through underlying physical laws. These properties are often hidden and cannot be directly observed, while the resulting system responses can only be measured through imperfect, often noisy and low-resolution, data. Accurately inferring these properties can reveal a system's internal characteristics and enable prediction of its response under new conditions. Governing physical laws provide a basis for inferring these properties from observed responses. However, this inverse problem is often severely ill-posed, exhibiting non-uniqueness, instability, or even nonexistence of solutions, with small observational errors potentially causing large errors in the inferred properties. These challenges are further exacerbated by the noisy and low-resolution data commonly available in practice. This talk will present our recent advances in physics-informed machine learning for statistical inverse problems, focusing on a unified probabilistic formulation that addresses these challenges, including ill-posedness, sensitivity to measurement noise, limited spatial resolution, error amplification through physics-based constraints, and learning the balance between data and physics.


**Biographical Sketch:** Dr. Jaesung Lee is an Assistant Professor in the Wm Michael Barnes '64 Department of Industrial and Systems Engineering at Texas A&M University. He received his Ph.D. in Industrial and Systems Engineering and M.S. in Statistics from the University of Wisconsin-Madison. His research focuses on domain-informed statistical machine learning incorporating uncertainty quantification, with applications in manufacturing processes and systems as well as biomedical domains. His research has been funded by the National Institutes of Health, the National Science Foundation, the U.S. Department of War, and industry partners. He has published articles in refereed journals, including Technometrics, IISE Transactions, and Advanced Science. He serves as an Associate Editor for the Journal of Intelligent Manufacturing. He received the IISE Transactions Best Paper Award, was a finalist in the 2025 INFORMS QSR Best Refereed Paper Competition, and was named a 2025-2026 Montague-Center for Teaching Excellence (CTE) Scholar at Texas A&M University.